Dimitris Selis

jsel32@icloud.com GitHub web LinkedIn Scholar

AI/ML Systems Researcher | Python, Kubernetes, Distributed Systems & MLOps
Cloud-native orchestration, autoscaling, observability and ML for systems.

EXPERIENCE

Athena Research & Innovation Center — Jan 2023 – Present, Athens, Greece
Collaborating Researcher

- Design and implementation of Python-based Kubernetes systems for autoscaling, scheduling, SLA-aware placement and elastic resource management.
- Implementation of reproducible cloud-native environments using Kubernetes, Docker, KEDA, Prometheus, Grafana/K6 and DeathStarBench to evaluate distributed workloads.
- Development of monitoring, observability and orchestration components for distributed AI/ML services operating under dynamic workloads, including anomaly detection, concept-drift monitoring, model-performance evaluation and automated remediation workflows.
- Technical work from architecture and implementation through debugging, experimentation, documentation and cross-team delivery.

Freelance Programming Tutor — Mar 2022 – Feb 2024, Remote

- Tutored university students in similarity-based information retrieval, search systems and distributed data lookup. Assisted in responsive web development, providing conceptual guidance, debugging support and code reviews.

Saphetor — Sep 2021 – Mar 2022, Lausanne, Switzerland
Software Engineer / Research Intern

- Developed Python backend services and reproducible ML pipelines for genomic anomaly detection.

PLATFORM SYSTEMS & OPEN SOURCE

KEDA External Scaler — Open Source
Implemented a gRPC-based external scaler for Kubernetes Event-Driven Autoscaling, enabling custom metric-driven scaling for distributed workloads.

SCALER — Self-consistent SLA Assurance and Remediation — IEEE ICC 2026
Co-built a Kubernetes-native SLA assurance pipeline combining autoencoder-based anomaly detection with Mistral-7B reasoning and SBERT semantic-similarity scoring for trustworthy explanations and semantic-drift detection.

MIND — Kubernetes-Native Scheduler and Placement Optimizer — IEEE TCC
Built a cloud-native orchestration system combining Transformer-based workload forecasting and reinforcement-learning scheduling for SLA-aware service placement.

KARMA — Knowledge-Aware Autoscaling for Cloud-Edge Workloads — IEEE GLOBECOM 2025
Developed a learning-based autoscaling system for SLA-aware resource allocation across distributed cloud-edge workloads.

ADDAPT6G — Distributed Cloud-Edge CNF Orchestration — IEEE ICC 2025
Built and evaluated a multi-domain cloud-edge testbed using LXD containers and Docker, integrating distributed RL agents to monitor resources and orchestrate SLA-aware service migration.

TECHNICAL STACK

Programming & Systems
Python · Go · C++ · Bash · JavaScript · Linux

Cloud-Native & Distributed Systems
Docker · Kubernetes · KEDA · Prometheus · Grafana · K6 · Istio · Kiali · GKE

Machine Learning Systems
PyTorch · StableBaselines3 · TensorFlow · Transformers · Hugging Face

MLOps & Data Systems
Kubeflow · Weights & Biases · SageMaker · PostgreSQL · InfluxDB · Qdrant

EDUCATION

Polytechnic University of Catalonia (UPC) Jan 2023 – Dec 2026 (expected)
PhD in Signals Theory and Communications Barcelona, Spain
Research: ML-driven resource management, distributed scheduling, autoscaling, reinforcement learning, observability.

University of Patras Oct 2014 – Sep 2021
MEng in Computer Engineering Patras, Greece
Thesis: Designed and deployed a containerized weather-forecasting platform on GKE.
Technical stack: Go · JavaScript · Docker · Kubernetes · Prometheus · Grafana

SELECTED PUBLICATIONS

- **D. Selis**, K. Ramantas, L. Alonso and C. Verikoukis, "MIND the Proximity: A Transformer-based RL Scheduler and Placement Optimizer for Cloud-edge Workloads, *IEEE Transactions on Cloud Computing*, under review.
- Z. Panou, **D. Selis**, K. Ramantas, L. Alonso and C. Verikoukis, "SCALER: Self-Consistent Assessment for SLA Explanation and Resolution in 6G Networks," *IEEE ICC 2026*
- **D. Selis**, K. Ramantas, L. Alonso, J. S. Vardakas and C. Verikoukis, "KARMA: Knowledge Aware Resource Management and Auto-scaling for cloud-edge workloads", *IEEE GLOBECOM 2025*
- **D. Selis**, K. Ramantas, L. Alonso, J. S. Vardakas and C. Verikoukis, "ADDAPT6G: Advanced Double Dueling Architecture for Proactive Tuning in 6G Networks," *IEEE ICC 2025*

AWARDS

IEEE CSIM 6G Industrial Dissemination Day Award 2025
Recognized for originality, technical rigor and industry relevance. Travel sponsorship for IEEE ICC 2025

TEACHING & MENTORING

Teaching Assistant 2024, 2025, 2026
5G Architectures, Enabling Technologies and KPIs University of Patras

Teaching Assistant Fall 2024
Fundamentals of Cellular Networks University of Patras

MEng Computer Engineering Student Advisor 2024–Present
Supervised research on offline RL and RL-assisted elastic service management for SLA-aware cloud-edge systems. University of Patras

Early-Stage Researcher Supervisor 2024-Present
Supervise ML systems research, guiding research design, implementation, experimentation, evaluation and technical writing.

LANGUAGES

Greek Native
English Fluent
French Upper Intermediate